

TRACK AND TRACE SUPPLYCHAIN

A PROJECT REPORT

Submitted by,

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Under the guidance of,

Ms. AYESHA TARANUM

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

At



**GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS**

PRESIDENCY UNIVERSITY

BENGALURU

DECEMBER 2024

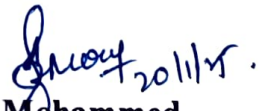
PRESIDENCY UNIVERSITY

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CERTIFICATE

This is to certify that the Project report “TRACK AND TRACE SUPPLYCHAIN” being submitted by “K V BRAMHANANDA REDDY” bearing roll number(s) “20211CSE0148”, by “THOTA YASWANTH RAM” bearing roll number(s) “20211CSE0093”, by “KUMMARA CHARAN SAI” bearing roll number(s) “20211CSE0097”, by “D GURUVARDHAN REDDY” bearing roll number(s) “20211CSE0121” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** is a bonafide work carried out under my supervision.


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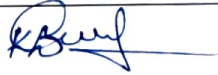
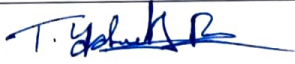
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DECLARATION

We hereby declare that the work, which is being presented in the project report entitled **TRACK AND TRACE SUPPLYCHAIN** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering**, is a record of our own investigations carried under the guidance of Ms. Ayesha Taranum, Assistant Professor, School of Computer Science Engineering & Information Science, Presidency University, Bengaluru.

We have not submitted the matter presented in this report anywhere for the award of any other Degree.

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ABSTRACT

In today's interconnected global economy, the complexity of supply chains has grown exponentially, involving multiple stakeholders, diverse geographies, and intricate processes. These complexities, coupled with the increasing demand for transparency and efficiency, have highlighted the need for a robust Track and Trace Supply Chain System. This project is designed to address these challenges by integrating advanced technologies such as the Internet of Things (IoT), Radio Frequency Identification (RFID), GPS tracking, blockchain, and artificial intelligence (AI). Together, these tools enable seamless tracking and tracing of goods from their point of origin to their final destination, ensuring transparency, accountability, and operational efficiency throughout the supply chain.

Track and trace systems have emerged as an indispensable component of modern supply chain management, addressing the growing demand for transparency, efficiency, and security. These systems enable businesses to monitor and document the movement of goods at every stage of their journey, from raw material sourcing to final delivery. By utilizing technologies such as barcodes, QR codes, RFID tags, IoT sensors, GPS, and blockchain, track and trace systems offer real-time visibility and ensure accountability. They are particularly valuable in industries where quality, compliance, and safety are critical, such as pharmaceuticals, food, automotive, and retail.

The system leverages IoT and RFID technology to continuously monitor and record critical parameters such as temperature, humidity, and location. This ensures that goods are maintained under optimal conditions, especially in sensitive industries like pharmaceuticals and food. GPS tracking combined with geo-fencing provides real-time location updates, alerts for unauthorized deviations, and route optimization. Blockchain technology plays a vital role in maintaining data integrity by creating immutable and tamper-proof records, fostering trust among stakeholders. Smart contracts further automate critical processes such as payments and compliance checks, reducing the likelihood of disputes and delays.

Global supply chains have become increasingly complex due to factors like globalization, outsourcing, and diverse regulatory requirements. This complexity creates risks such as counterfeiting, theft, inefficiencies, and product recalls. Track and trace systems address these challenges by enabling businesses to track the current location of products and trace their history in case of issues. They also support compliance with stringent regulations, improve customer trust, and optimize logistics.

AI and machine learning are at the core of the system's predictive capabilities. By analyzing historical data and current trends, these technologies provide insights into potential risks and recommend proactive measures to mitigate them.